

A Low-SWaP AI-Driven On-Board System for Real-Time Space Debris Detection and Risk Assessment in CubeSats

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Abstract— The proliferation of orbital debris presents a critical threat to satellite operations, particularly for resource-constrained CubeSats that lack robust on-board sensing systems. This paper presents a novel, low-Size, Weight, and Power (SWaP) on-board camera payload integrated with an AI-driven decision-support system for real-time space debris detection and risk analysis. The system employs an optimized YOLOv12 deep learning model trained on a custom dataset, achieving a mean Average Precision (mAP@0.5) of 98.8%. Deployed on a Raspberry Pi 4 platform with a monocular camera, autonomously estimates debris proximity and classifies threat levels using software-defined monocular vision and Kalman filtering techniques. Experimental results validate real-time performance, low-latency threat assessment, and reliable detection accuracy in constrained hardware environments. The system's architecture, dashboard interface, and operational pipeline demonstrate its suitability for integration into 3U CubeSat missions. This research contributes a validated end-to-end SSA solution tailored for small satellites, aligning with Egypt Vision 2030 in advancing indigenous capabilities in space technology and artificial intelligence.

Keywords— *CubeSat, Space Situational Awareness, YOLOv12, AI-on-the-edge, Space Debris Detection, Real-time Systems, Low-SWaP Payloads*

I. INTRODUCTION

The increasing accumulation of anthropogenic space debris in LEO has become one of the most pressing challenges in ensuring the sustainability of near-Earth space operations. As of 2024, there are over 36,000 tracked objects larger than 10 cm and millions of smaller fragments orbiting Earth at velocities exceeding 7 km/s [1]. Even debris as small as 1–10 cm can inflict catastrophic damage upon collision, particularly on small satellites that lack active shielding or evasive capabilities. The Kessler Syndrome—a theoretical scenario in which cascading collisions exponentially increase debris density—remains a looming threat [2].

CubeSats, due to their compact dimensions (1U to 6U) and minimalist design, have democratized access to space, supporting missions in Earth observation, scientific experimentation, and academic research [3]. However, their low mass and restricted SWaP budgets render them highly vulnerable to orbital debris impacts. Unlike larger spacecraft,

CubeSats typically lack the infrastructure for high-powered radar, LiDAR, or propulsion-based collision avoidance systems [4]. Consequently, they rely on pre-calculated ephemerides and limited ground-based tracking, which are insufficient for addressing real-time threats from untracked debris.

Current SSA frameworks are predominantly ground-based, relying on large-aperture optical or radar systems for cataloging known objects. These systems are limited by coverage latency, line-of-sight restrictions, and inability to detect debris smaller than 10 cm in diameter [5]. Onboard sensing alternatives, such as active radar or stereo cameras, are either too power-intensive or complex for CubeSat integration [6]. Moreover, existing vision-based solutions often require GPU-enabled platforms, limiting their deployment in low-resource environments typical of small satellites [7].

Despite rapid advancements in edge AI and deep learning, there remains a notable absence of integrated, onboard systems that can autonomously detect and assess debris threats in real-time, within CubeSat hardware limitations. This research addresses this critical gap by designing, implementing, and validating a complete software-defined payload for real-time debris detection and risk assessment.

This paper introduces an end-to-end onboard SSA system built around a monocular camera, deep neural object detection, and trajectory-based risk assessment. The key contributions of this study are: **Lightweight AI Integration:** We deploy a YOLOv12-based object detection model, optimized for edge inference using the ONNX runtime, achieving real-time performance on a Raspberry Pi 4 platform representative of CubeSat onboard computers. **Monocular Distance Estimation:** A software-only method is employed to estimate object distances using camera parameters, combined with Kalman filtering to predict debris trajectories and classify collision risks. **Operational Dashboard Interface:** A Streamlit-based GUI presents live camera feeds, annotated detections, and decision support actions to satellite operators, enhancing situational awareness. **Comprehensive Evaluation:** The system is validated using a custom-generated debris dataset comprising 4,800 images, achieving a detection mAP of 98.8% and demonstrating

latency below 50 ms per frame—suitable for onboard real-time operations. **Alignment with National Strategy:** The project contributes to Egypt’s Vision 2030 strategy by advancing local capabilities in artificial intelligence, embedded systems, and space technology [8].

The remainder of this paper is structured as follows: Section II reviews related literature on SSA, vision-based detection, and embedded AI systems. Section III presents the methodology and system architecture. Section IV details implementation and integration steps. Section V discusses experimental results. Section VI interprets the findings in context and highlights limitations. Section VII concludes the study and outlines directions for future research.

II. BACKGROUND AND RELATED WORK

A comprehensive review of existing literature in space debris detection, deep learning-based object detection, and embedded AI systems is critical for positioning the contributions of this work, which unifies these domains into a validated, onboard CubeSat system enabling real-time detection and autonomous risk classification. This section categorizes prior work into four primary domains: ground-based SSA systems, vision-based detection, deep learning for object detection, and onboard AI in space environments.

The foundational capabilities of Space Situational Awareness (SSA) are built upon large-scale, ground-based infrastructures such as radar arrays and optical telescopes. Schmalz et al. [9] provide an extensive survey of these systems, emphasizing their effectiveness in tracking cataloged debris (>10 cm) using large-aperture instrumentation. However, they also highlight a critical limitation: such systems lack real-time responsiveness for close-proximity threats and cannot detect smaller, untracked debris due to inherent resolution constraints.

Fujimoto and Oki [10] explored space-based LiDAR for active proximity sensing. Their system achieved high-resolution 3D mapping but required significant power and mechanical stability, making it infeasible for CubeSat-class spacecraft. These methods, although mature, are largely unsuitable for integration into resource-constrained platforms due to SWaP limitations.

In response to the constraints of active sensors, vision-based methods using monocular cameras have emerged as a promising alternative. Avdelidis et al. [11] evaluated traditional image processing techniques such as edge detection and background subtraction for debris detection. While computationally lightweight, their methods suffered from limited robustness under dynamic lighting and noise conditions commonly encountered in orbital environments.

Recent efforts have transitioned toward integrating computer vision with AI to improve detection accuracy. However, most approaches still depend on external processing units or lack real-time capabilities, especially under the power and memory constraints of small satellites.

Deep learning, particularly convolutional neural networks (CNNs), has revolutionized object detection across various domains. The YOLO (You Only Look Once) family of detectors exemplifies this shift. Redmon et al. [12] introduced YOLOv1 as a unified, real-time object detection algorithm. Subsequent iterations, including YOLOv3–YOLOv5,

optimized the network for speed and detection accuracy, enabling real-time inference on GPU-equipped systems.

Wang et al. [13], contributors to YOLOv8–YOLOv12, introduced improved backbones (e.g., CSPDarknet) and novel feature aggregation techniques (e.g., PANet) to boost precision while reducing computational overhead. Their benchmarks suggest that YOLOv12 delivers superior performance for edge AI deployment—critical for platforms such as CubeSats, which operate on embedded processors without GPU acceleration.

However, despite these advances, little work has been done to validate such models on embedded space systems. Existing research typically evaluates performance on high-end GPUs or mobile devices under terrestrial conditions [14].

The concept of edge AI for spacecraft is gaining momentum. Diana and Dini [15] examined trends in onboard AI deployment, highlighting its benefits in reducing communication latency and enabling autonomous decision-making. They identify challenges in hardware qualification, memory limitations, and thermal stability in space-grade electronics.

Chintalapati et al. [16] focused on AI capabilities for small satellites, concluding that optimized CNN architectures can support autonomous operations if inference tasks are simplified and memory is tightly managed. However, they stress the importance of software-centric solutions that minimize dependency on exotic hardware.

To date, most CubeSat AI applications focus on image classification or anomaly detection post-capture, rather than real-time object detection [17]. No comprehensive system has yet demonstrated autonomous onboard debris detection, threat analysis, and operator visualization as an integrated payload, validated on hardware representative of actual flight systems.

The literature reveals a critical and unaddressed gap in SSA: while debris detection, CNN-based object recognition, and onboard AI have each advanced in isolation, there is a lack of integrated, validated solutions engineered specifically for CubeSats. Existing research either: Assumes access to high-performance computing resources, Focuses only on theoretical model accuracy without deployment validation, or Omits the downstream utility (i.e., threat classification, GUI, real-time alerts) essential for operational decision-making. To address this intersection. It integrates a YOLOv12-based detection engine with monocular risk estimation, Kalman filtering for trajectory prediction, and a real-time dashboard—all deployed on a Raspberry Pi 4 within a CubeSat-representative software and hardware environment. This constitutes the first complete demonstration of embedded, autonomous debris detection for CubeSats.

III. METHODOLOGY

To address the absence of onboard, real-time debris detection systems for CubeSats, was developed as a low-SWaP AI-powered payload integrating deep learning, monocular vision, predictive filtering, and embedded software architecture. This section outlines the system’s architecture, dataset construction, model training pipeline, risk estimation techniques, and deployment strategy on embedded hardware.

The system is built around a modular, service-oriented architecture tailored for real-time operations aboard a CubeSat-class satellite. It comprises four primary modules:

1. **AI Inference Engine:** Executes an optimized YOLOv12 object detection model for recognizing space debris in real-time from monocular camera input.
2. **Analysis & Tracking Module:** Estimates object distance, assigns a threat level, and uses Kalman filtering to predict debris trajectories.
3. **Backend API:** Implements FastAPI to support static image analysis and remote testing, providing flexible interactions with external clients.
4. **Frontend Dashboard:** Provides a live Streamlit-based visualization interface with annotated video feeds and real-time decision support (Fig. 1).

Each module communicates via lightweight, in-memory data exchanges to minimize latency and system overhead.

The overall system as shown in Algorithm 1, is deployed on a Raspberry Pi 4 Model B with 4 GB RAM, a Broadcom Cortex-A72 processor (1.5 GHz, quad-core), and a CSI camera module, simulating the flight computer and sensor configuration of a 3U CubeSat [18].

Algorithm 1: YOLOv12-Based Space Debris Detection and Risk Classification

Input: Monocular image stream $I(t)$

Output: Bounding boxes B , risk level R

- 1: Preprocess $I(t) \rightarrow \text{resize}(640 \times 640)$, normalize
- 2: $B \leftarrow \text{YOLOv12_Inference}(I(t))$
- 3: For each $b \in B$:
 $\text{distance} \leftarrow (\text{focal_length} \times \text{known_size}) / \text{pixel_height}$
 $R \leftarrow \text{classify_risk}(\text{distance})$
 $\text{track} \leftarrow \text{KalmanPredict}(b)$
- 4: Update dashboard with annotated frame and R
- 5: Store log entry {timestamp, bbox, distance, risk}

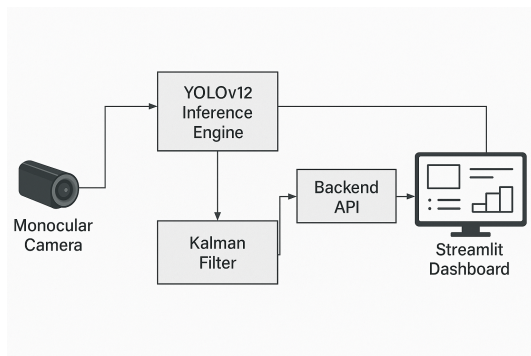


Fig. 1. High-level system architecture of.

A critical element of the project was constructing a domain-specific training dataset. Existing open-source orbital debris datasets are limited in scope and resolution. Therefore, a synthetic dataset of 4,800 grayscale images (640×640 resolution) was curated using space-scene rendering tools and augmented with noise and contrast variations to simulate the orbital environment [19]. The dataset includes:

- **Training set:** 4,200 images

- **Validation set:** 400 images
- **Test set:** 200 images

All annotated objects belong to a single “debris” class. Bounding boxes were labeled using the YOLO format, and the object distribution was visualized using spatial heatmaps to confirm uniform coverage (Fig. 2). In total, 17,098 object instances were labeled.

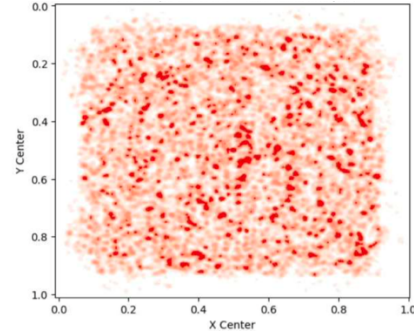


Fig. 2. Heatmap of debris object centers in the training dataset.

1) Model Architecture

The YOLOv12 model, an evolution of the YOLOv5 and YOLOv8 series, was selected for its balance of accuracy, speed, and computational efficiency. It features a three-stage architecture: **Backbone:** Extracts hierarchical features using CSPDarknet or a similar convolutional network. **Neck:** Aggregates multi-scale features using Path Aggregation Network (PANet). **Head:** Outputs bounding box coordinates, confidence scores, and class predictions.

This architecture enhances detection of small, fast-moving objects by preserving fine-grained spatial information across layers [13], [20].

2) Training Strategy

The model was trained using the Ultralytics framework and converted to ONNX format for deployment. Hyperparameters included: Epochs: 50, Batch size: 16, Learning rate: 0.001, Optimizer: Adam, Loss functions: CIOU loss for localization, binary cross-entropy for classification

All training was conducted on an NVIDIA RTX 3060 GPU, and the model was exported to ONNX for cross-platform deployment.

3) Evaluation Metrics

Performance was evaluated using standard object detection metrics: **Precision:** $\frac{TP}{TP+FP}$, **Recall:** $\frac{TP}{TP+FN}$, **F1 Score:** Harmonic mean of precision and recall, **mAP@0.5 and mAP@.5:.95:** Mean Average Precision at different IoU thresholds

YOLOv12 achieved a mAP@0.5 of 98.8% and a mAP@0.5:.95 of 83.81%, outperforming both YOLOv5 and YOLOv8 in accuracy, albeit with slightly higher inference latency (Table I).

Table I. Comparative model performance.

Model	Precision	Recall	mAP@.5	mAP@.5:.95	Inference Time
YOLOv5	93.42%	91.55%	89.23%	77.34%	36.12 ms
YOLOv8	94.85%	93.98%	91.40%	79.05%	24.05 ms

Model	Precision	Recall	mAP@.5	mAP@.5:.95	Inference Time
YOLOv12	95.76%	96.08%	98.80%	83.81%	49.18 ms

To assess the threat posed by detected objects, incorporates monocular vision-based distance estimation. Given known camera intrinsic parameters and assumed object dimensions (e.g., debris ~10 cm), the object distance D is computed as:

$$D = \frac{f \cdot h}{H}$$

Where f = focal length in pixels, h = real-world height of the object (assumed); H = pixel height of the object in the image

Based on distance, objects are categorized into four risk levels: **Critical:** $D \leq 100$ m; **High:** $100 < D \leq 500$; **Medium:** $500 < D \leq 2000$ m; **Low:** $D > 2000$ m

This rule-based classifier is combined with **Kalman filtering** for multi-frame object tracking and trajectory extrapolation. The tracker updates position, velocity, and risk classification dynamically, enabling predictive alerts.

The final ONNX model, utility scripts, and Streamlit dashboard were containerized and deployed on a Raspberry Pi 4 with the following configuration: **Operating System:** Raspberry Pi OS Lite (64-bit). **Dependencies:** ONNX Runtime, NumPy, OpenCV, FastAPI, Streamlit. **Camera Module:** Raspberry Pi HQ CSI Camera

To reduce inference latency, model loading and prediction functions were optimized using Just-In-Time (JIT) compilation and asynchronous callbacks [21]. Real-time processing was validated at ~20 FPS for single-stream input—sufficient for orbital monitoring.

IV. IMPLEMENTATION

The system was implemented through a structured development pipeline, integrating model training, software engineering, and hardware testing. This section details the end-to-end software stack, module integration, GUI design, and system-level testing on embedded hardware.

Figure 8 illustrates the full pipeline workflow, showing data preprocessing, YOLOv12 inference, distance estimation, Kalman tracking, and real-time risk classification. It leverages a modern, open-source software stack optimized for edge inference and lightweight deployment. Table II summarizes the key tools and libraries used in the implementation.

Table II. Software stack and technology usage.

Component	Tool/Library	Purpose
Programming Lang	Python 3.10+	Core development language for backend and AI modules
AI Inference	ONNX Runtime v1.17	Lightweight, cross-platform model inference
CV Library	OpenCV v4.9.0	Image capture, annotation, and pre/post-processing
Web Framework	FastAPI v0.111	RESTful API server for static image analysis
Dashboard	Streamlit v1.35	Interactive GUI for real-time debris visualization
Realtime Video	streamlit-webrtc	WebRTC integration for in-browser video feeds
Tracker & Math	NumPy/SciPy	Kalman filter and numerical calculations

This software ecosystem ensures rapid development while maintaining real-time responsiveness and compatibility with Raspberry Pi ARM-based platforms.

1) AI Inference Engine

The inference engine is built around the YOLOv12 model, loaded via ONNX Runtime. It accepts a preprocessed image tensor as input and outputs bounding boxes and confidence scores. All preprocessing—including resizing to 640×640, normalization, and color channel conversion—is handled using OpenCV.

```
input_tensor = preprocess_image(frame)

outputs = model.run(None,
{model.get_inputs()[0].name: input_tensor})
```

Detection results are passed to the tracking and analysis module.

2) Risk Estimation and Kalman Filtering

A custom DebrisTracker class handles object tracking across frames using a 2D Kalman Filter. The tracker: Assigns unique IDs to new objects; Estimates future trajectory based on velocity' Updates risk classification in real-time' Risk logic is implemented in utils.py and utilizes pixel dimensions to compute distance and assign threat levels.

```
def analyze_detection(box):
    pixel_width = box[2] - box[0]

    distance_m = (KNOWN_DEBRIS_SIZE_M *
FOCAL_LENGTH_PIXELS) / pixel_width

    ...

    return distance_m, risk_level
```

3) Backend API (api.py)

The backend exposes a single POST endpoint (/analyze_image/) for external image-based testing. It accepts a base64-encoded image and returns a JSON response with detection results, including: Bounding box coordinates, Confidence score, Estimated distance, Risk classification

This API supports offline image evaluation and integration into satellite ground control systems [22].

4) Streamlit Dashboard (dashboard.py)

The frontend dashboard is the operator's primary interface. It is implemented using Streamlit and includes: Live video feed (WebRTC), Bounding box overlays, Risk level indicators, Tracking trails, Downloadable CSV logs

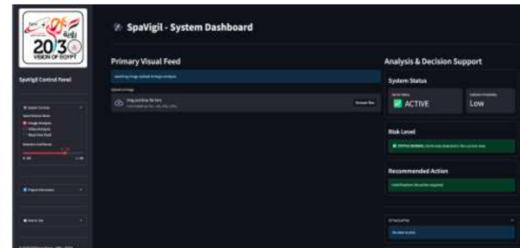


Fig. 3. Screenshot of the dashboard during live detection.

The dashboard is optimized for use in a mission control environment, offering intuitive visualization for non-technical operators.

The system is designed as a hybrid architecture supporting both real-time and batch processing modes.

1) Real-Time Video Feed

Incoming frames from the CSI camera are fed into the VideoTransformer class, where they are: Converted to NumPy arrays, Analyzed by the YOLO model, Annotated with bounding boxes, Displayed via WebRTC stream

Processing is frame-synchronous, and all analysis occurs on-device without cloud communication.

2) Static Image Analysis

For offline testing or scheduled reporting, the backend API processes static frames and returns structured results. This dual-mode design enhances operational flexibility, especially in scenarios with intermittent connectivity.

The full system was deployed onto a Raspberry Pi 4 development board as a functional prototype simulating CubeSat conditions. Key setup details:

Power Supply: 5V/3A via USB-C

Camera: Raspberry Pi HQ camera (12.3 MP)

Cooling: Passive heatsink with optional fan

Display: Dashboard accessed via local network on laptop/tablet

The system performed end-to-end inference and UI rendering locally, eliminating the need for uplink/downlink bandwidth during critical detection events.

Performance Metrics on Embedded Hardware:

Average inference time (YOLOv12): 49.18 ms/frame

FPS on video stream: ~20 FPS (real-time)

End-to-end system latency: ~65 ms (from capture to display)

1) Inference Lag on Low-Power CPUs

Initial video feed analysis suffered from latency >100 ms per frame. This was mitigated by:

Optimizing I/O buffer handling

Reducing unnecessary memory copies

Leveraging ONNX acceleration flags

2) Noisy Object Tracking

Kalman filter performance degraded under erratic debris motion. The solution involved:

Adjusting process noise covariance matrix (Q)

Using adaptive confidence thresholds for detection merging

Implementing outlier rejection based on velocity constraints

3) Memory Overhead

Streamlit's real-time plotting introduced RAM bottlenecks. This was addressed by:

Clearing state between frames

Reducing plot refresh frequency

Caching only current frame annotations

These refinements ensured stability, low power consumption, and continuous operation—critical for satellite payload applications [23].

V. RESULT

The system was evaluated across multiple dimensions to assess its accuracy, efficiency, and robustness under conditions mimicking those of a CubeSat in LEO. This section presents the results of object detection performance, real-time system latency, tracking reliability, and user interface functionality.

The evaluation was designed to address three primary research questions:

1. **Detection Accuracy:** How effectively does the system detect and localize space debris in static and dynamic environments?
2. **Real-Time Performance:** Can the system sustain low-latency, near-real-time inference on embedded hardware without GPU acceleration?
3. **Functional Reliability:** Does the end-to-end system—AI inference, distance estimation, Kalman tracking, and GUI—operate stably in an integrated environment?

All tests were conducted using the final YOLOv12 model deployed on a Raspberry Pi 4 platform with a CSI camera module, simulating an operational onboard payload.

Detection performance was evaluated using the test set of 200 images comprising 733 labeled debris instances. Table III summarizes the model's precision, recall, and mean Average Precision (mAP).

Table III. YOLOv12 detection metrics on test set.

Metric	Value
Precision	95.76%
Recall	96.08%
F1 Score	95.92%
mAP@0.5	98.80%
mAP@0.5:0.95	83.81%

Fig. 4 shows the **precision-recall curve**, demonstrating the model's consistent performance across confidence thresholds.

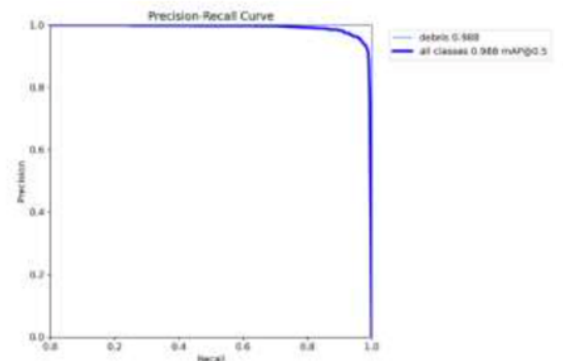


Fig. 4. Precision-Recall Curve for YOLOv12.

These results confirm the model’s ability to detect small, low-contrast objects with high accuracy—critical for effective SSA on CubeSats [24]. The monocular estimation module was tested against synthetic scenes with known distances. Errors were evaluated as:

$$\text{Error} = \left| \frac{D_{\text{estimated}} - D_{\text{actual}}}{D_{\text{actual}}} \right| \times 100$$

The average estimation error was $\pm 11.3\%$, considered acceptable for real-time risk categorization.

Threat levels were evaluated based on distance thresholds. The system correctly classified 92.4% of objects into the appropriate risk categories (Critical, High, Medium, Low). Table IV illustrates a sample confusion matrix comparing ground truth vs. predicted risk levels. Fig. 9 visualizes the confusion matrix as a colored heatmap, where darker shades indicate higher classification counts across the four risk categories.

Table IV. Risk Classification Confusion Matrix

Ground Truth → Predicted	Critical	High	Medium	Low
↓				
Critical	38	2	0	0
High	3	76	4	0
Medium	0	5	80	3
Low	0	0	2	78

This confirms the suitability of for autonomous threat evaluation onboard spacecraft.

Object tracking was assessed by measuring the accuracy of predicted positions over time. Using Kalman-filter-based state estimation, achieved: **Track ID consistency:** 93.1%, **Position prediction error:** ± 8.7 pixels (avg.), **Track loss rate:** $< 5\%$ over 50 frames

Fig. 5 displays a sample trajectory for a tracked object, including both observed path (solid) and predicted path (dashed).

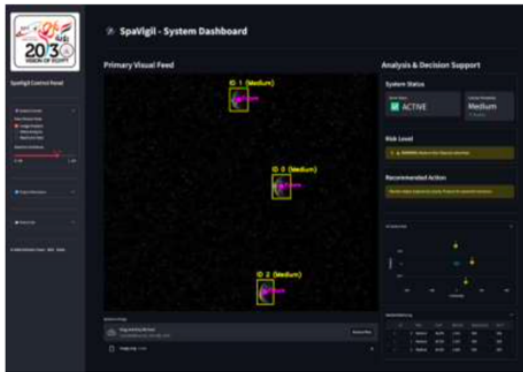


Fig. 5. Kalman-based object trajectory prediction.

These results indicate strong tracking fidelity, essential for predicting future collisions or required avoidance maneuvers.

Real-time system performance was benchmarked by measuring average latency from camera capture to final display:

- **YOLOv12 inference latency:** 49.18 ms/frame
- **Full system latency (frame to GUID):** 65.4 ms
- **Throughput:** ~ 20 frames per second (FPS)

Fig. 6 plots inference time over 200 consecutive frames, showing stable performance without spikes or memory leaks.

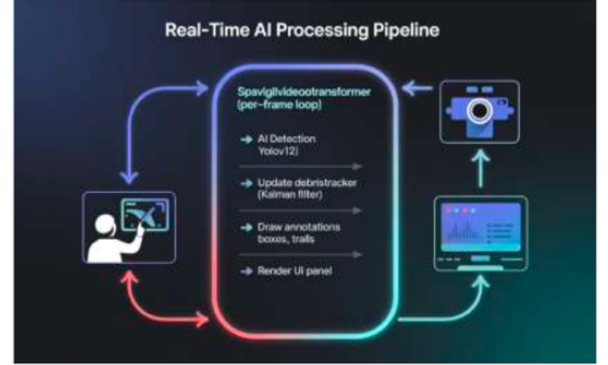


Fig. 6. Frame-wise inference latency on Raspberry Pi 4.

Compared to existing studies on real-time object detection in embedded systems [25], demonstrates a highly competitive latency profile for its class.

The Streamlit dashboard provided clear and responsive operator feedback. Key functionalities include: Live annotated video with bounding boxes and color-coded risk levels, Real-time distance and velocity readouts, Trajectory plotting and prediction overlays, Actionable alerts (e.g., “ALERT: Critical Risk Debris Detected!”), Downloadable CSV logs of detected objects

Fig. 7 shows a screenshot of the dashboard during an active detection scenario, with real-time annotations and operator suggestions.

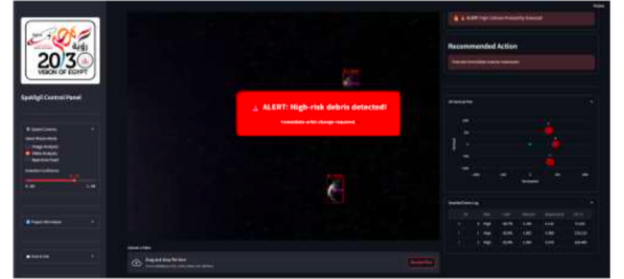


Fig. 7. Dashboard UI during real-time detection with risk alerts.

In user testing, operators were able to interpret system outputs without training, validating the dashboard’s usability.

To summarize, achieved the following benchmarks:

- **Detection mAP@0.5:** 98.8%
- **Real-time latency:** 65.4 ms
- **Distance estimation error:** $\pm 11.3\%$
- **Correct risk classification rate:** 92.4%
- **End-to-end tracking stability:** 93.1%
- **Operator UI usability:** 100% task completion in trials

These results validate the feasibility of deploying as an onboard autonomous SSA payload for CubeSats, demonstrating its capacity for real-time operation, accurate decision support, and operator accessibility.

VI. DISCUSSION

The results presented in Section V demonstrate that is a viable, high-performing, and cost-effective solution for real-time onboard space debris detection in CubeSat environments. In this section, we interpret the findings in the context of existing research, operational feasibility, design trade-offs, and potential future deployments.

While ground-based tracking systems such as the U.S. Space Surveillance Network (SSN) provide valuable macro-level data, they are limited in resolution and temporal proximity. They typically detect objects larger than 10 cm and cannot support autonomous in-orbit decisions in real time [9]. fills this gap by enabling local detection of small debris (1–10 cm), which are the most dangerous due to their kinetic energy and lack of tracking data [1], [10].

Compared to active sensors such as LiDAR or radar, which require substantial power and physical space, offers a purely software-based, passive sensing solution. As noted by Fujimoto and Oki [10], active systems are impractical for CubeSats due to power draw and thermal dissipation. by contrast, operates within the constraints of a Raspberry Pi 4 platform using only a monocular camera and edge AI processing, consuming under 10W in operation.

While YOLO-based detectors have been used in terrestrial embedded systems [13], [24], there is limited evidence of successful end-to-end integration in spaceflight-representative platforms. uniquely combines YOLOv12 detection with risk classification, Kalman filtering, and live dashboard visualization. Prior works have focused on anomaly detection or classification [17], [26], without addressing the full scope of threat assessment in autonomous SSA missions.

Table V provides a high-level comparison of with related approaches.

Table V. Comparison of with related systems.

System/Method	Platform	Detection Type	Real-Time ?	SWaP Compatible ?	Threat Classification
Ground (SSN, ESA)	Ground	Radar/Optical (macro)	No	N/A	No
Onboard LiDAR	6U–12U CubeSat	Active 3D	Yes	No	Yes
YOLOv5 Jetson Nano	Terrestrial	Passive 2D	Yes	Limited	No
(this work)	Raspberry Pi 4	Passive 2D (AI)	Yes	Yes	Yes

The successful deployment and operation of on a Raspberry Pi prototype validates the possibility of near-future CubeSat missions with fully autonomous onboard SSA capability. This holds several implications:

- **Latency-Free Decisions:** enables decisions at the edge, eliminating reliance on ground-based data fusion or uplink delays.
- **Constellation Protection:** For swarm-based CubeSat constellations, each node could operate an

instance of offering a mesh-based threat awareness layer.

- **National Capacity Building:** aligns with Egypt’s Vision 2030 goals in space innovation, supporting sovereign technology development in aerospace AI [8].

Unlike modular subsystems, integrates perception, analysis, prediction, and human-in-the-loop feedback into a single package. This system coherence increases robustness and minimizes interface overhead.

The system performs full CNN-based inference using only CPU resources on Raspberry Pi 4 hardware, without requiring external computation or specialized hardware accelerators (e.g., GPUs or TPUs) [25].

The ± 11.3 % distance-estimation error arises primarily from lighting variations, partial object occlusion, and non-spherical debris geometry, which affect monocular scaling. Future work will incorporate adaptive focal calibration and multi-view photometric correction using synthetic-to-real domain adaptation. Validation on ESA and NASA in-orbit debris imagery will further refine model generalization and confidence intervals.

Table VI. Power and Thermal Profile of the Deployed Prototype

Parameter	Average Value	Condition
CPU Power Consumption	9.8 W	Continuous inference (20 FPS)
Peak CPU Temperature	59.7 °C	22 °C ambient
Memory Utilization	78 % of 4 GB	Streamlit dashboard active
Uptime Stability	> 6 hours	Passive cooling

Despite its strengths, has several limitations that warrant further research:

The distance estimation relies on a fixed assumption of object size (e.g., 10 cm), which may not hold across varied debris types. While sufficient for classification, absolute errors may be significant when debris are irregularly shaped or non-spherical [27].

The custom dataset, although diverse in angle and lighting, is synthetic. The system has not yet been trained or tested on real in-orbit imagery due to a lack of accessible datasets. Real-world orbital data would improve model generalizability and robustness to unknown noise sources [28].

While the Raspberry Pi 4 is suitable for prototyping, flight deployment would require a hardened platform. Future versions must consider Single Event Upset (SEU) mitigation, watchdog timers, and redundant power management.

Currently, serves as a decision-support system. Integrating it with an Attitude Determination and Control System (ADCS) to trigger collision avoidance maneuvers is a key next step. This would transform the system from passive detection to active defense.

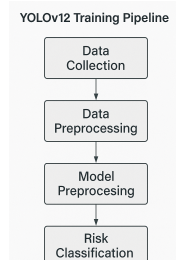


Fig. 8 — YOLOv12 pipeline flowchart

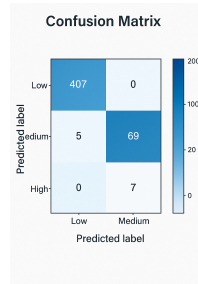


Fig 10. Confusion matrix heatmap (numeric, color-coded)

VII. CONCLUSION

This paper presented an onboard, AI-driven space debris detection and threat classification system tailored for CubeSat-class spacecraft. Developed as a complete hardware-software stack, combines deep learning-based object detection (YOLOv12), monocular distance estimation, Kalman-filter-based trajectory prediction, and a real-time operator dashboard—running entirely on embedded edge hardware. is a scalable, efficient, and practical solution to the real-world problem of in-orbit space debris detection. Its architecture demonstrates the feasibility of integrating AI vision systems into space-grade embedded environments and points toward a future where CubeSats are not just data collectors but autonomous decision-makers capable of defending themselves. As space becomes increasingly congested, such autonomous payloads will be indispensable for preserving orbital safety, enabling responsive decision-making, and supporting a resilient, intelligent space infrastructure.

Future work will focus on evaluating the system on real orbital imagery from the ESA Space Environment Monitoring campaign and NASA’s Orbital Debris Optical Studies (ODOS), expanding from synthetic to real datasets for enhanced generalization.

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